

Module-1

Introduction to financial management: Meaning and objectives of Financial Management, changing role of finance managers. Interface of Financial Management with other functional areas.

Indian Financial System: Capital Market, Money Market, Forex Market, Commodity market financial services.

Emerging issues in Finance: Crypto currency, Financial Engineering, Behavioral finance (Theory).

Introduction to Finance

At the macro level, finance is the study of financial institutions and financial markets and how they operate within the financial system in both the Indian and global economies.

At the micro level, finance is the study of financial planning, asset management, and fund raising for businesses and financial institutions.

Meaning of Financial Management

Financial management is concerned with the **acquisition, financing, and management of assets with some** overall goal in mind.

In other words, financial management involves the efficient use of financial resources in the production of goods.

Financial Management is concerned with three activities:

- (i) **Anticipate financial needs:** - which mean estimation of funds required.
- (ii) **Acquiring financial resources:** - i.e., where and how to obtain the funds to finance the anticipated financial needs.
- (iii) **Allocating funds in business:** - allocation of available funds among the best plans, which are able to maximize shareholder's wealth.

Thus, the decisions of financial management can be divided into three viz., investment, financing and dividend decisions.

Definitions of Financial Management

- ➔ **According to Wheeler** defines financial management as “the business activity which is concerned with the acquisition & conservation of capital funds in meeting the financial needs & overall objectives of business enterprise”
- **According to Guttman and Dougall:** financial management is defined “as the activity concerned with the planning, raising, controlling and administering the funds used in the business.”

Scope or Contents of Financial Management

1. Estimating Financial requirements
2. Deciding Capital Structure
3. Selecting Sources of finance
4. Selecting pattern of investment
5. Proper cash management
6. Implementing Financial controls
7. Proper use of surpluses

Objectives of Financial Management

1. Profit maximization
2. Wealth maximization

1. Profit maximization

Profit is a primary motivating force for any economic activity. Firm is essentially being an economic organization, it has to maximize the interest of its stakeholder. The firm has to earn profit from its operations. Profit cannot be ignored since it is both a measure of the success of business and the means of its survival and growth. Profit is the positive and fruitful difference between revenues and expenses of a business enterprise over a period of time.

- If an enterprise fails to make profit, capital invested is eroded and if this situation prolongs, the enterprise ultimately ceases to exist. The overall objective of business enterprise is to earn at least satisfactory returns on the funds invested, consistent with maintaining a sound financial position.

- As applicable to financial management, the profit maximization criterion implies that the investment, financing & dividend policy decisions of a firm should be oriented to the maximisation of profit / EPS
- Maximizing the Rupee Income of Firm
 - ➔ Resources are efficiently utilized
 - ➔ Appropriate measure of firm performance
 - ➔ Serves interest of society also
- **The rationale/ arguments behind profit maximization**
 - (i) It is a test of economic efficiency
 - (ii) Provides yardstick to judge the economic performance
 - (iii) Leads to efficient allocation of resources
 - (iv) Ensures maximum social welfare
- ➔ **The Drawbacks/Limitations of the objective of profit maximization**
 - (i) Ambiguity and vagueness of the concept of profit
 - (ii) It ignores the timing of the returns or benefits
 - (iii) The quality of benefits is ignored i.e., the risk is ignored
 - (iv) Ignores Perfect Competition
 - (v) Ignores dividend policy

2. Wealth maximization

This is also known as **value maximisation or net present worth maximisation** of the shareholder wealth maximization (shareholders' value maximization) is also a main objective of financial management. Wealth maximization means to earn maximum wealth for the shareholder. So, the finance manager tries to give a maximum dividend to the shareholder. He also tries to increase the market value of the shares.

The market value of the shares is directly related to the performance of the company. Better the performance, higher is the market value of shares and vice-versa. So, the finance manager must try to maximize shareholder's value.

- **Argument in favor of Wealth maximization**
 - (i) It serves the interests of the owners as well as other stakeholders
 - (ii) It is consistent with the objective of owner's economic welfare
 - (iii) It implies long – run survival & growth of the firm
 - (iv) It effects on the dividend policy
 - (v) It considers risk factor & time value of money

➤ **Criticism/ Drawbacks of wealth maximization**

- (i) It is a prescriptive but not descriptive and does not explain of what firms actually do.
- (ii) The objective of wealth maximisation is not necessarily socially desirable
- (iii) There is some controversy whether the aim to maximise the wealth of the firm or the shareholder
- (iv) It also faces some difficulties when ownership and management are separated as in the case of large corporations.

Roles of Finance Managers

1. Traditional Role

2. Changing Role / Emerging Role

1. Traditional role of finance managers

- 1. In the past the role of financial manager was passive in other decision making.
- 2. He was considered as a part of the accounts department.
- 3. His role was to maintain the store records, preparing accounting and other quasi-financial reports, raising funds when needed.
- 4. He was playing a passive role of an advisor.
- 5. He was an office staff.

2. Changing role of finance managers

- 1. Financial forecasting and planning
- 2. Acquisition of funds
- 3. Investment of funds
- 4. Helping in valuation decisions
- 5. Maintain proper liquidity
- 6. Profit planning
- 7. Understanding capital markets

Interface of Financial Management with Other functional areas**1. Financial Management and Production Management**

Production management is the operational part of the business concern, which helps to multiple the money into profit. Profit of the concern depends upon the production performance. Production performance needs finance, because production department requires raw material, machinery, wages, operating expenses etc. These expenditures are decided and estimated by the financial department and the finance manager allocates the appropriate finance to production department. The financial manager must be aware of the operational process and finance required for each process of production activities.

2. Financial Management and Economics

Economic concepts like micro and macroeconomics are directly applied with the financial management approaches. Investment decisions, micro and macro environmental factors are closely associated with the functions of financial manager. Financial management also uses the economic equations like money value discount factor, economic order quantity etc. Financial economics is one of the emerging area, which provides immense opportunities to finance, and economical areas.

3. Financial Management and Accounting

Accounting records includes the financial information of the business concern. Hence, we can easily understand the relationship between the financial management and accounting.

In the olden periods, both financial management and accounting are treated as a same discipline and then it has been merged as Management accounting because this part is very much helpful to finance manager to take decisions. But nowadays financial management and accounting discipline is separate and interrelated.

4. Financial Management and Mathematics

Modern approaches of the financial management applied large number of mathematical and statistical tools and techniques. They are also called as econometrics. Economic order quantity, discount factor, time value of money, present value of money, cost of capital, capital structure theories, dividend theories, ratio analysis and working capital analysis are used as mathematical and statistical tools and techniques in the field of financial management.

5. Financial Management and Marketing

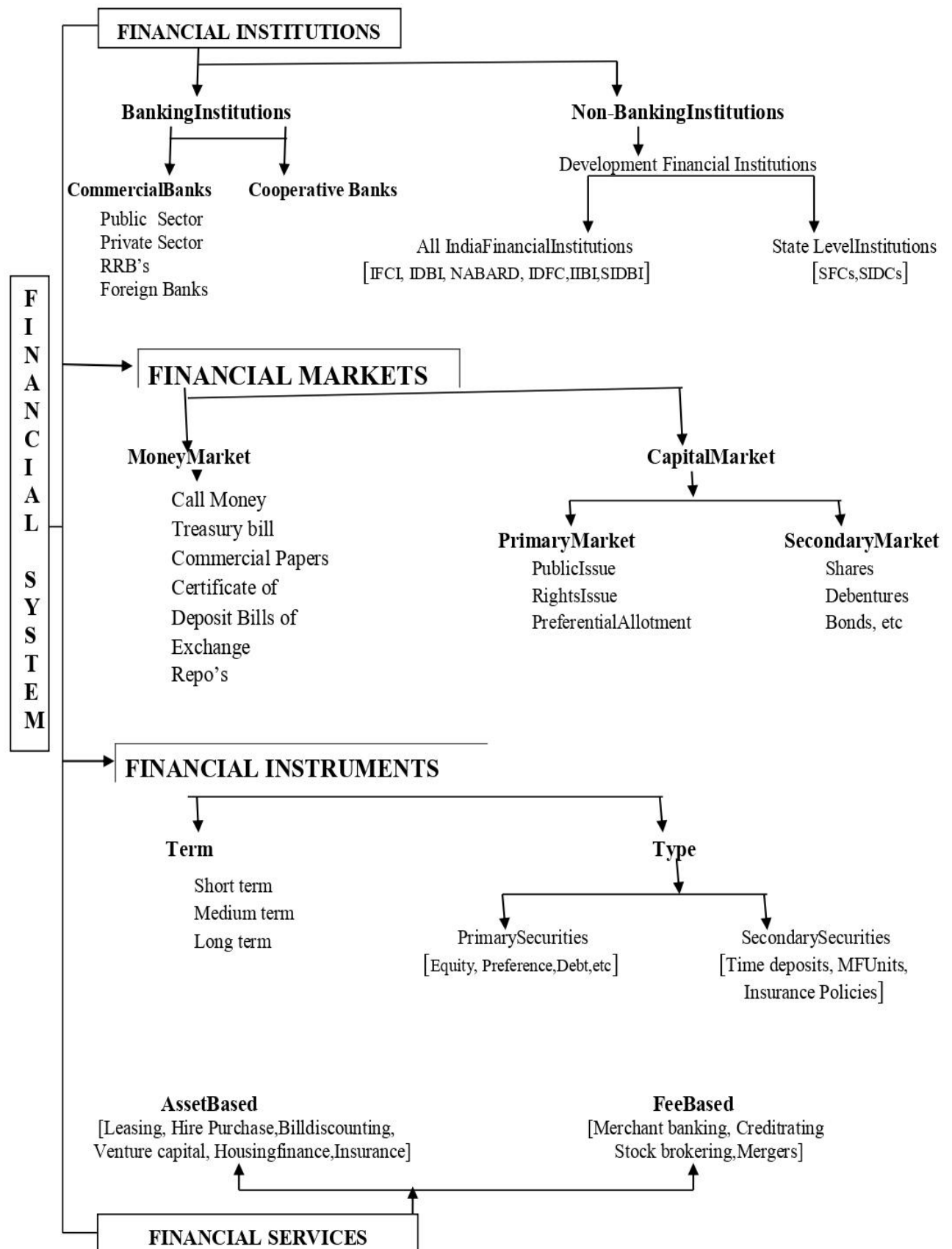
Produced goods are sold in the market with innovative and modern approaches. For this, the marketing department needs finance to meet their requirements. The financial manager or finance department is responsible to allocate the adequate finance to the marketing department. Hence, marketing and financial management are interrelated and depends on each other.

6. Financial Management and Human Resource

Financial management is also related with human resource department, which provides manpower to all the functional areas of the management. Financial manager should carefully evaluate the requirement of manpower to each department and allocate the finance to the human resource department as wages, salary, remuneration, commission, bonus, pension and other monetary benefits to the human resource department. Hence, financial management is directly, related with human resource management.

Indian Financial System

Indian Financial System comprises of financial institutions, financial markets, financial instruments and the services provided by the financial institutions for a cost. It provides an avenue by which savings are transformed into investments. Financial system plays an important role in the functioning of the economy because it facilitates transfer of economic resources from one section of the economy to another. Finance is the study of money, its acquisition, allocation, management and regulations, therefore financial system includes all these activities dealing in finance, organized into a system.



I. Financial Institutions

In its simple meaning a financial institution is an institution that provides financial services. Financial services refer to the activity of transfer of financial resources from the savers or investors to the users or borrower. Thus, essentially financial institutions are the intermediaries between the savers and borrowers of the money. More specifically, the term “financial institutions” refers to all kinds of organizations which intermediate and facilitate financial transactions of both individual and corporate customer.

→ Banking Institutions

According to the Banking Regulations Act, a banking institution is an institution that does the business of banking. The term banking business is defined as the accepting of deposits of money from the public for the purpose of lending or investment and repayable on demand. A banking institution mobilizes the savings of the public through accepting of deposits of money and lends the same to the individual and corporate customers to meet their short term, medium term and long term financial requirements and invests the surplus amount in various securities or financial instruments in the financial market.

• Commercial banks

The banks which engage in the activity of providing finance to various commercial establishments for meeting their working capital requirements are called as commercial banks. The term “working capital requirement” refers to the funds required for meeting the day to day activities like buying of raw materials, payment of salaries, etc. These banks raise the required funds in the form various kinds of deposits from the public and lend short term and medium-term loans to traders and business persons mainly for meeting working capital requirements.

• Co-operative Banks

The banks which are formed on the principle of mutual co-operation to meet the needs of its members are called as co-operative banks. They are formed by small group of individuals belonging to same or similar kind of activities like farming, retailing, small scale industry, etc. These banks raise the required funds in the form of various kinds of deposits from the members and the public. They lend short term and medium-term loans to members and other traders, business persons and agriculturists mainly for meeting working capital requirements at generally low rate of interest and play a significant role in financing agricultural and allied activities in India.

➔ Non-Banking Financial Institution

A Non-Banking Financial Institution popularly called as Non-Banking Financial Company (NBFC) refers to a specialized financial institution other than a banking institution that mobilizes the savings of a certain segment of the public and lends the same to the individual and corporate customers to meet their specialized needs. It is registered under the Companies Act, and is engaged in the business of lending to specialized purposes and investing in shares, debentures, bonds, etc.

As per section 45(1)(f) of the Reserve Bank of India Act, a non-banking financial company is a non-banking institution which is a company and which has its principal business the receiving of deposits under any scheme or lending in any manner. It is compulsory for a NBFC to get itself registered with RBI as a deposit accepting company. A company intending to do the business as a NBFC must have a minimum of ₹ 25 lakhs as owned fund. It should have more than 50% of assets in the form of financial assets and more than 50% of income as income from financial assets.

• Development Financial Institutions

They are the institutions that provide long term finance for agriculture and industrial development purposes. These are multipurpose institutions that provide not only credit facility but also assist in discovering investment projects, preparing project reports, arranging technical advice, managing industrial units, underwriting, promotional activities, etc. Basically they are intended to develop backward regions as well as small and new entrepreneur. They provide medium and long term finance to business units mainly for acquisition / development of basic facilities like land and buildings, plant and machinery, etc. These institutions generally do not accept deposits from the public. IDBI, ICICI, IFCI, SIDBI, SFCs & SIDC fall under this category.

II. Financial Markets

Money always flows from surplus sector to deficit sector. That means persons having excess of money lend it to those who need money to fulfill their requirement. Similarly, in business sectors the surplus money flows from the investors or lenders to the businessmen. So, we find two different groups, one who invest money or lend money and the others, who borrow or use the money.

The financial markets act as a link between these two different groups. It facilitates this function

by acting as an intermediary between the borrowers and lenders of money. So, financial market may be defined as ‘a transmission mechanism between investors (or lenders) and the borrowers (or users) through which transfer of funds is facilitated’. It consists of individual investors, financial institutions and other intermediaries who are linked by a formal trading rules and communication network for trading the various financial assets and credit Instruments.

➔ **Money market:**

Money market is a place where instruments with high liquidity and very short-term maturities are traded. It refers to an activity, which involves borrowing and lending of short term funds which have a maturity period of up to one year.

Money Market Instruments:

1. Treasury Bills

Treasury bills are the short term promissory note issued by the RBI on behalf of the government, in order to meet the temporary deficit of the government. They are issued with three-month, six-month and one-year maturity periods which may also be referred as 91- day, 182-day and 364-day treasury bills.

2. Commercial Papers

It is a short term unsecured promissory note issued by reputed companies, issued at a discounted value on face value. They are usually issued with fixed maturity between 1 to 270 days and for financing of accounts receivables, inventories and meeting short term liabilities. It is a low-cost alternative to bank loans.

3. Certificate of Deposit

It is a short term borrowing more like a bank term deposit account. It is a promissory note issued by a bank in the form of a certificate entitling the bearer to receive interest. The certificate bears the maturity date, the fixed rate of interest and the value. Its term generally ranges from three months to five years and restricts the holders to withdraw funds on demand. However, on payment of certain penalty the money can be withdrawn on demand also.

4. Repurchase transactions, called Repo or Reverse Repo

They are transactions or short-term loans in which two parties agree to sell and repurchase the same security. Under repurchase agreement the seller sells specified securities with an agreement to repurchase the same at a mutually decided future date and price. Similarly, the buyer purchases the securities with an agreement to resell the same to the seller on an agreed date at a predetermined price.

Such a transaction is called a Repo when viewed from the perspective of the seller of the securities and Reverse Repo when viewed from the perspective of the buyer of the securities.

5. Commercial Bills:

It is a short-term credit investment created by a non-financial firm and guaranteed by a bank to make payment. It is simply a bill of exchange drawn by a person and accepted by a bank. It is a buyer's promise to pay to the seller a certain specified amount at certain date. The same is guaranteed by the banker of the buyer in exchange for a claim on the goods as collateral. The most common term for these instruments is 90 days. However, they can vary from 30 days to 180 days. The seller need not hold it until maturity and can sell off the same in secondary market at discount from the face value to liquidate its receivables.

→ Capital Market:

Capital market is generally referred to as the market for long term funds. It includes institutions and mechanism for the effective pooling of long term funds from individuals and institutional investors and making them available to industrial and commercial undertakings. Capital market, in short, deals with shares, debentures, bonds etc. It is an imperative in economic development as it mobilizes funds on a large scale and distributes the same to places where it is needed in an economy.

- **Primary market/New issue market**

The primary market deals with the new securities which were not previously available to the investing public i.e., the securities that are offered for the first time to the investing public. The market therefore makes available a new block of securities for public subscription. Initial public offer (IPO) (in case of an unlisted company), Follow-on public offer (FPO), rights issue and bonus issue are some of the types of issues in this market.

- **Secondary Market/Stock Market**

It is a market for old securities or second-hand securities, where existing securities are traded. It is called as secondary market or stock market. The securities which are already issued in the primary market, is traded in the secondary market. The functions of the secondary market are basically done through a stock exchange. Stock exchange is a market which facilitates the trading of the capital market instruments which are mentioned below.

Capital Market Instruments – some of the capital market instruments are:

1. Shares
2. Debenture/ Bonds
3. Derivatives

Difference between capital market and money market

Sl No	Capital Market	Money Market
1.	It provides finance/money for long-term investment.	It provides finance/money for short-term investment.
2.	The finance provided by the capital market may be used both for fixed and working capital.	The finance provided by money market is utilized, usually for working capital.
3.	Stock Exchange acts as an investment market for buyers and sellers of securities.	It does not provide such facilities. The main components include call loan market, collateral loan market, and bill market and acceptance houses.
4.	It acts as a middleman between the investor and the entrepreneur.	It acts as a link between the depositor and the borrower.
5.	Underwriting is one of its primary activities.	Underwriting is a secondary function.
6.	It provides long-term funds to Central and State Governments, public and local bodies for development purposes.	It provides short-term funds to government by purchasing treasury bills and to others by discounting bills of exchange, etc.

Difference between Primary market and Secondary market

Sl No	Primary Market	Secondary Market
1	It deals only with new or fresh issue	It deals in existing securities
2	No fixed geographical location needed	Needs a fixed place for market activities
3	Securities are transferred from corporate to investors	Securities are transferred from one investor to investors
4	Listing of securities is not required	Only listed securities are traded here
5	Prices are determined by the management	Prices of securities are determined by the market forces such as demand and supply

- **Commodities Market:**

A physical or virtual marketplace for buying and selling raw or primary products. Commodities are split into two types: hard and soft commodities. Hard commodities are typically natural resources that must be mined or extracted (gold, silver, rubber, oil, etc.), whereas soft commodities are agricultural products or livestock (corn, wheat, coffee, sugar, soybeans, pork, etc.)

- **Forex Markets:**

Foreign exchange market provides the physical and institutional structure through which the money of one country is exchanged for that of another country, the rate of exchange between currencies is determined, and foreign exchange transactions are physically completed. A

foreign exchange transaction is an agreement between a buyer and a seller that a given amount of one currency is to be delivered at a specified rate for some other currency

III. Financial Instruments

Another important constituent of financial system is financial assets/instruments. They represent a claim against the future income and wealth of other in other words, a financial instrument is a claim, against a person or an institution, for the payment of a sum of money and/or a periodic payment in the form of interest or dividend, at a specified future date. In financial markets, a variety of investors operate. To suit their requirements, different types of securities are issued by the companies and financial institutions. In other words, financial markets/system promotes development of innovative financial products suited to the "investment requirements of heterogeneous investor.

Financial securities may be classified under two broad categories

1. Primary Securities.

These are also termed as 'direct securities' as these are issued directly by the ultimate borrowers of funds to the ultimate savers or investor Primary securities include equity shares, preference shares and debentures.

2. Secondary Securities.

These securities are also termed as 'indirect securities' as these are not issued directly by the ultimate borrowers, rather, these are issued by financial intermediaries to ultimate saver Insurance policies, units of the mutual funds, bank deposits etc. are the examples of secondary securities.

IV. Financial Services

The term financial services can be defined as "activities, benefits and satisfactions, connected with the sale of money, that offer to users and customers, financial related value".

Fund Based Services Include

1. Equipment Leasing/Finance
2. Hire-Purchase and Consumer Credit
3. Bill Discounting
4. Venture Capital
5. Housing Finance
6. Insurance Services
7. Factoring etc.

Fee Based/Advisory Services Include

1. Issue Management
2. Portfolio Management
3. Corporate Counseling
4. Loan Syndication
5. Merger and Acquisition
6. Capital Restructuring
7. Credit Rating, Stock Broking etc.,

Emerging Issues in Financial Management**1. Financial Engineering**

Financial engineering is the use of **mathematical techniques** to solve financial problems. Financial engineering uses tools and knowledge from the fields of computer science, statistics, economics, and applied mathematics to address current financial issues as well as to devise new and innovative financial products.

Financial engineers design, create and implement new financial instruments, models & processes to solve problems in finance and take advantage of new financial opportunities.

Financial Engineering aims at the following

- Designing & developing new financial instruments
- Formulating new process
- Formulating creative solutions to financial problems

2. Behavioral Finance

Behavioral finance, a sub-field of behavioral economics, proposes psychology-based theories to explain stock market anomalies, such as severe rises or falls in stock price.

Behavioral finance, with its roots in the psychological study of human decision-making, is a relatively new and evolving subject in the field of finance. In brief, behavioral finance is the study of investors' psychology while making investment decisions.

The purpose is to identify and understand why people make certain financial choices. Within behavioral finance, it is assumed the information structure and the characteristics of market. Participants systematically influence individuals' investment decisions as well as market outcomes.

Behavior finance focuses upon how investors interpret and act on information to make

informed investment decisions. Investors do not always behave in a rational, predictable and an unbiased manner indicated by the quantitative models. Behavioral Finance places an emphasis upon investor behavior leading to various market anomalies.

Question Paper Questions**3 Marks**

1. What is financial management?
2. What is money market?
3. What are the objectives of financial management?
4. Name the two aims of finance function.
5. What are the 3 functions of financial management?
6. Define financial engineering.
7. Mention any three importance of financial management.
8. State the three types of financial decisions.
9. What is crypto currency
10. What do mean by behavioral finance?
11. Write a note on financial engineering.

7 & 10 Marks

1. Explain the interface of financial management with different functional areas.
2. Explain Indian financial system.
3. Wealth maximization objective is superior to profit maximization, explain.
4. Briefly explain the role of finance manager in current volatile environment.
5. Briefly explain the scope of financial management.
6. Distinguish between capital market and money market
7. Briefly explain the functions of financial management.
8. What are the arguments in favor of profit maximization?
9. Write a note on capital market
10. Differentiate between primary market and secondary market.
11. Discuss the recent issues in financial Management
12. What are the primary objectives of financial management? Explain
13. Explain how wealth maximization is better objective is better than profit maximization for business today.
14. Wealth maximization objective is superior to profit maximization objective. Do you agree?
15. Briefly explain the changing role of financial managers
16. Describe the functions of primary and secondary market
17. What is money market? Briefly explain money market instruments

MODULE-2**Time Value of Money**

The value of unit of money is different in different time period. It is also called as time preference for money.

In other word an individual's preference for possession of a given amount of cash now rather than the same amount at some future time, is called time value of money.

Techniques of time value of money

- I. Compounding Technique (Future Value Technique)
- II. Discounting Technique (Present Value Technique)

I. Compounding Technique (Future Value Technique)

Under this technique interest is compounded when the amount earned on an initial deposit (principal amount) becomes the part of the principal amount at the end of the first compounding period that will be continued up to the estimated future period.

In other words, Interest is compounded when the amount earned on an initial deposit (the initial principal) becomes part of the principal at the end of the first compounding period.

The term **principal** refers to the **amount of money on which interest is received**.

Methods of Compounding Technique (Future Value Technique)

- 1. Future Value of Single cash flow
 - 1.1 Annually
 - 1.2 Semi-annually & other compounding period
 - 1.3 Inter compounding period
- 2. Compounding Value of Annuity
- 3. Doubling Period
- 4. Effective Rate of Interest (ERI)

1. Future value of a single amount / cash flow

⇒ **Annually**

$$\text{Compounding Value (Future Value)} = \text{Present Value} (1+r)^n$$

⇒ **Semi Annually & other compounding period**

$$\text{Compounding Value (Future Value)} = \text{Present Value} \left[1 + \left(\frac{r}{m} \right) \right]^{m \cdot n}$$

r = Rate of interest

n = number of years

m = number of times compounded. So, m = A-1, SA-2, M-12, BM-6

PROBLEMS

1. Deposit ₹15,000 per year in a bank, which provides interest @ 11% p.a., compounded half yearly. What will be the amount received by him after 6 years?

$$\text{Compounding Value (Future Value)} = \text{Present Value} \left[1 + \left(\frac{r}{m} \right) \right]^{m \cdot n}$$

$$\text{Compounding Value (Future Value)} = ₹15,000 \left[1 + \left(\frac{0.11}{2} \right) \right]^{2 \cdot 6}$$

$$\text{Compounding Value (Future Value)} = ₹15,000 [1 + (0.055)]^{12}$$

$$\text{Compounding Value (Future Value)} = ₹15,000 [1.9012]$$

$$\text{Compounding Value (Future Value)} = ₹28,518$$

2. **Future value or Compound Value of an Annuity**

⇒ **If deposit at the end of the year**

$$\text{Future Value} = PV_1(1+r)^{n-1} + PV_2(1+r)^{n-2} + PV_3(1+r)^{n-3} \dots\dots\dots + PV_n(1+r)^{n-n}$$

(OR)

$$\text{Future Value} = \text{Present Value (FVIFA } r, n)$$

⇒ **If deposited beginning of the year**

$$\text{Future Value} = PV_1(1+r)^n + PV_2(1+r)^{n-1} + PV_3(1+r)^{n-2} \dots\dots\dots$$

2. Mr. 'X' deposits ₹ 2,000 at the end of every year for 5 years in his savings account paying 5% interest compounded annually. He wants to determine how much he will have at the end of 5th year.

Solution

$$\text{Future Value} = PV_1(1+r)^{n-1} + PV_2(1+r)^{n-2} + PV_3(1+r)^{n-3} \dots\dots\dots + PV_n(1+r)^{n-n}$$

$$\begin{aligned} \text{Future Value} &= ₹2000(1+0.05)^{5-1} + ₹2000(1+0.05)^{5-2} + ₹2000(1+0.05)^{5-3} + ₹2000(1+0.05)^{5-4} \\ &+ ₹2000(1+0.05)^{5-5} \end{aligned}$$

$$\text{Future Value} = ₹2431.01 + ₹2315.25 + ₹2205 + ₹2100 + ₹2000$$

$$\text{Future Value} = ₹11051.26$$

3. Doubling period**(i) Rule 72**

$$= \frac{72}{\text{Rate of Interest}}$$

(ii) Rule 69

$$0.35 + \frac{69}{\text{Rate of Interest}}$$

{Note: - Interest rate should take whole value}

3. Effective Rate of Interest

Effective rate of interest is when the frequency of interest is compounded more than once in a year, then interest amount added to the principal amount for the next year.

$$\text{ERI} = \left[1 + \left(\frac{r}{m} \right) \right]^m - 1$$

3. Mr. Baba deposited ₹ 1,000 in a bank at 10% rate of interest with quarterly compounding. He wants to know the effective interest rate.

Solution: -

$$\text{ERI} = \left[1 + \left(\frac{r}{m} \right) \right]^m - 1$$

$$\text{ERI} = \left[1 + \left(\frac{0.1}{4} \right) \right]^4 - 1$$

$$\text{ERI} = 0.1038 = 10.38\%$$

II. Present Value Technique (OR) Discounting Technique

Present value technique shows what the today's value is of future sum of money. Present value is the concept that states an amount of money today is worth more than that same amount in the future. In other words, money received in the future is not worth as much as an equal amount received today.

Methods of Present Value Technique (OR) Discounting Technique

1. Present Value of Single Cash Flow
2. Present Value of Annuity
3. Present Value of infinite period **or** continuous period

⇒ **Present value of Single cash flow / Single amount**

$$\text{Present Value} = \text{Future Value} \left[\frac{1}{1+r} \right]^n$$

(OR) – Table Method – A3 table

$$\text{Present Value} = \text{Future Value (PVF}_{r,n})$$

1. What is the present value of ₹ 1,00,000 to be received after 10 years, expected rate of return is 8%.

$$\text{Present Value} = \text{Future Value} \left[\frac{1}{1+r} \right]^n$$

$$\text{Present Value} = ₹1,00,000 \left[\frac{1}{1+0.08} \right]^{10}$$

$$\text{Present Value} = ₹ 46,319$$

(OR) – Table Method – A3 table

$$\text{Present Value} = \text{Future Value (PVF}_{8\%, 10 \text{ years})}$$

$$\text{Present Value} = ₹1,00,000 (0.463_{8\%, 10 \text{ years}})$$

$$\text{Present Value} = ₹46,300$$

⇒ **Present value of uneven series (or) Annuity**

$$\text{Present Value} = \sum_{t=1}^n \text{FV}_1 \left[\frac{1}{1+r} \right]^{t1} + \text{FV}_2 \left[\frac{1}{1+r} \right]^{t2} + \text{FV}_3 \left[\frac{1}{1+r} \right]^{t3} \dots\dots\dots$$

(OR) – Table Method – A4 table-

$$\text{Present Value} = \text{Future Value (PVIFA}_{r,n})$$

2. Ms. Paru wants to determine the present value of annuity consisting cash inflows of ₹ 12,000 for 5 years & required ROI is 12%.

$$\text{Present Value} = \sum_{t=1}^5 \text{FV}_1 \left[\frac{1}{1+r} \right]^{t1} + \text{FV}_2 \left[\frac{1}{1+r} \right]^{t2} + \text{FV}_3 \left[\frac{1}{1+r} \right]^{t3} \dots\dots\dots$$

$$\text{Present Value} = \sum_{t=1}^5 12,000 \left[\frac{1}{1+0.12} \right]^1 + 12,000 \left[\frac{1}{1+0.12} \right]^2 + 12,000 \left[\frac{1}{1+0.12} \right]^3 + 12,000 \left[\frac{1}{1+0.12} \right]^4 + 12,000 \left[\frac{1}{1+0.12} \right]^5$$

$$\text{Present Value} = \sum_{t=1}^5 ₹ 10,714.29 + ₹9,566.4 + ₹ 8541.36 + ₹7626.22 + ₹ 6809.2$$

$$\text{Present Value} = ₹ 43,257.47$$

(OR) – Table Method – A4 table

$$\text{Present Value} = \text{Future Value (PVIFA}_{12\%, 5 \text{ years})}$$

$$\text{Present Value} = ₹12,000 (3.605_{12\%, 5 \text{ years}})$$

$$\text{Present Value} = ₹ 43,260$$

⇒ **Present value of Perpetuity (or) continues period**

$$\text{Present Value}_{\infty} = \frac{\text{Cash Inflow}}{\text{Interest Rate}}$$

1. Let us assume that an investor expects a perpetual sum of ₹ 5,000 annually from his investment. What is the value of his perpetuity if his interest rate is 10%?

$$\text{Present Value}_{\infty} = \frac{\text{Cash Inflow}}{\text{Interest Rate}}$$

$$\text{Present Value}_{\infty} = \frac{₹5,000}{0.10}$$

$$\text{Present Value}_{\infty} = ₹50,000$$

Application of Time Value of Money

- I. Sinking Fund Method
- II. Compound Growth Rate Method
- III. Interest Rate Method / Implicit Rate of Interest Method
- IV. Capital Recovery Method

I. Sinking Fund Method

A sinking fund is a fund containing money set aside or saved to pay off a debt or bond.

A financial manager wants to determine the annual amount has to be accumulating for a future date to replace the existing asset or to clear the liability.

$$\text{Sinking Fund} = \frac{V_n}{\text{ACF } r, n}$$

ACF -- A2 table

V_n = Value of fund required to maintain at the defined time period

1. A company has issued debentures of ₹ 50 lakhs to be repaid after 7 years. How much the company should invest in a sinking fund earning 12% in order to be able to repay debentures?

$$\text{Sinking Fund} = \frac{V_n}{\text{ACF } r, n}$$

$$\text{Sinking Fund} = \frac{₹50,00,000}{10.089_{12\%, 7 \text{ years}}}$$

$$\text{Sinking Fund} = ₹4,95,589.26$$

A company should invest ₹4,95,589.26 in a sinking fund in order to repay debenture at the end of 7th year.

II. Compound Growth Rate Method (CAGR)

Compound annual growth rate, or CAGR, is the annual growth rate of an investment over a specified period of time longer than one year.

$$V_n = V_o \{CF_{r,n}\}$$

V_n = Last or n^{th} year profit or investment or sales

V_o = Starting year profit or investment or sales

CF = Compound factor – **A1 table**

r = Rate of Interest

n = number of years

1. From the following data calculate compound growth rate in profit for XYZ Ltd.,

2016	2017	2018	2019	2020	2021
60	60	70	80	85	100

$$V_n = V_o \{CF_{r,n}\}$$

$$100 = 60 \{CF_{r,6 \text{ years}}\}$$

$$\{CF_{r,6 \text{ years}}\} = \frac{100}{60}$$

$$\{CF_{r,6 \text{ years}}\} = 1.667$$

$$r = 9\%$$

Using A1 (compound factor) table at the rate of 9% interest the compound value is 1.677. Therefore, the compound growth rate is 9%

III. Interest Rate Method / Implicit Rate of Interest Method

An implicit interest rate is an interest rate that is not specifically stated in a business transaction.

$$V_n = V_o \{ACF_{r,n}\}$$

V_n = Amount required to pay at end of the period

V_o = Amount required deposit annually

ACF = Compound factor – **A2 table**

r = Rate of Interest

n = number of years

1. A finance company advertises that it will pay a lump sum of ₹8,000 at the end of 6 years to investors who deposit annually ₹ 1,000 for 6 years. What interest rate is implicit in this offer?

$$V_n = V_o \{ACF_{r,n}\}$$

$$₹8,000 = ₹1,000 \{ACF_{r,6 \text{ years}}\}$$

$$ACF_{r,6 \text{ years}} = \frac{₹8,000}{₹1,000}$$

$$ACF_{r,6 \text{ years}} = 8$$

Using A2 (Annuity compound factor) table at the rate of 12% interest the annuity compound value is 8.115. Therefore, the implicit rate of interest in this offer is 12%

IV. Capital Recovery Method

$$\text{Loan Installement} = \left[\frac{\text{Loan Amount}}{PVIFA_{r,n}} \right]$$

1. A company raised a loan of ₹ 5,00,000 from a financial institution at 8% p.a. If amount has to be paid back in 15 installments. What shall be the installment size?

$$\text{Loan Installement} = \left[\frac{\text{Loan Amount}}{PVIFA_{r,n}} \right]$$

$$\text{Loan Installement} = \left[\frac{₹5,00,000}{8.560_{8\%,15 \text{ years}}} \right]$$

$$\text{Loan Installement} = ₹58411.21$$

Loan Amortization Schedule

The amortization schedule is a record of your loan payments that shows the principal amounts and the interest included in each payment. The schedule shows all payments until the end of the loan term.

1. Mr. Longmari borrow ₹10,00,000 at an interest rate of 15% and the loan is to be paid in 5 equal installments payable at the end of each of the next 5 years Prepare the loan amortization schedule.

Solution: -

$$\text{Loan Installement} = \left[\frac{\text{Loan Amount}}{\text{PVIFA}_{r,n}} \right]$$

$$\text{Loan Installement} = \left[\frac{₹10,00,000}{3.352_{15\%, 5 \text{ years}}} \right]$$

$$\text{Loan Installement} = ₹2,98,329$$

The loan amortization schedule of Mr. Longmari

No. of Installments	Beginning of loan amount	Installment Amount	Interest (cl2 *15%)	Principal Amount (cl3-cl4)	Ending of loan amount (cl2-cl5)
1	10,00,000	2,98,329	1,50,000	1,48,329	8,51,671
2	8,51,671	2,98,329	1,27,751	1,70,578	6,81,093
3	6,81,093	2,98,329	1,02,164	1,96,165	4,84,928
4	4,84,928	2,98,329	72,739	2,25,590	2,59,338
5	2,59,338	2,98,329	38,901	2,59,338	0

Module -3

Sources of Capital & Cost of Capital

COST OF DEBENTURES OR BONDS

- (i) Irredeemable or perpetual debenture
- (ii) Redeemable debenture

Irredeemable or perpetual debenture

This type of debenture usually issued for a lifetime of a company. Once the company issues irredeemable debenture or bonds it will exist in the company for lifetime.

$$K_{d \text{ bt}} = \left[\frac{\text{Interest}}{\text{Net Proceeds}} \right]$$

$$K_{d \text{ at}} = \left[\frac{\text{Interest}}{\text{Net Proceeds}} \right] * [1 - t]$$

Net Proceeds = Amount received @ the time of issue after adjusting premium, discount & any other charges

1. (i) XYZ takes ₹ 56000, 8% debenture at par. The tax rate is 35%.
- (ii) PQ co. issues ₹ 80000, 10% debentures at premium of 10%. The tax rate is 60%
- (iii) ABC company issues ₹ 60,000, 8% debentures at discount of 5%. The tax rate is 50%
- (iv) MN Ltd issues ₹ 1,20,000, 9% debentures at premium of 10%. Cost of flotation is 3% & tax rate is 65%

Calculate cost of debt for the above cases.

Solution: -

$$K_{d \text{ bt}} = \left[\frac{\text{Interest}}{\text{Net Proceeds}} \right]$$

$$K_{d \text{ at}} = \left[\frac{\text{Interest}}{\text{Net Proceeds}} \right] * [1 - t]$$

- (i) NP= ₹56,000 I= ₹56,000*8%=₹4480 t=35%

⇒ Cost of debt before tax

$$Kd_{bt} = \left[\frac{\text{Interest}}{\text{Net Proceeds}} \right]$$

$$Kd_{bt} = \left[\frac{₹4,480}{₹56,000} \right]$$

$$Kd_{bt} = 0.08 = 8\%$$

⇒ Cost of debt After tax

$$Kd_{at} = \left[\frac{\text{Interest}}{\text{Net Proceeds}} \right] * [1 - t]$$

$$Kd_{at} = \left[\frac{₹4,480}{₹56,000} \right] * [1 - 0.35]$$

$$Kd_{at} = 0.052 = 5.2\%$$

- (ii) NP = Issue Price + Premium = ₹80,000 + (₹80,000 * 10%) = ₹88,000
 I = ₹80,000 * 10% = ₹8,000
 t = 60%

$$Kd_{bt} = 9.09\%$$

$$Kd_{at} = 3.64\%$$

- (iii) NP = Issue Price - Discount = ₹60,000 - (₹60,000 * 5%) = ₹57,000
 I = ₹60,000 * 8% = ₹4,800
 t = 50%

$$Kd_{bt} = 0.0842 = 8.42\%$$

$$Kd_{at} = 0.042 = 4.2\%$$

- (iv) NP = Issue Price + Premium - ((Issue Price + Premium) * Floatation cost)
 = ₹1,20,000 + (₹1,20,000 * 10%) - (₹1,32,000 * 3%)
 = ₹1,32,000 - ₹3,960
 = ₹1,28,040

$$I = ₹1,20,000 * 9\% = ₹10,800$$

$$t = 65\%$$

$$Kd_{bt} = 0.0843 = 8.43\%$$

$$Kd_{at} = 0.0295 = 2.95\%$$

REDEEMABLE DEBENTURE (OR) BONDS

Redeemable debentures are those debentures which can be redeemed after certain period of time.

⇒ **Before Tax**

$$Kd_{bt} = \frac{\text{Interest} + \frac{1}{n} (\text{Redeemable Value} - \text{Net Proceeds})}{\frac{1}{2} (\text{Redeemable Value} + \text{Net Proceeds})}$$

⇒ **After Tax**

$$Kd_{at} = \frac{\text{Interest}(1-t) + \frac{1}{n} (\text{Redeemable Value} - \text{Net Proceeds})}{\frac{1}{2} (\text{Redeemable Value} + \text{Net Proceeds})}$$

2. A company issues ₹12 lakh at 10% redeemable debentures at a discount of 5%. The cost of flotation is ₹ 35000. The debentures are redeemable after 5 years. Calculate before & after tax of debt assuming tax rate of 50%.

⇒ **Before Tax**

$$Kd_{bt} = \frac{\text{Interest} + \frac{1}{n} (\text{Redeemable Value} - \text{Net Proceeds})}{\frac{1}{2} (\text{Redeemable Value} + \text{Net Proceeds})}$$

$$Kd_{bt} = \frac{₹1,20,000 + \frac{1}{5} (₹12,00,000 - ₹11,05,000)}{\frac{1}{2} (₹12,00,000 + ₹11,05,000)}$$

$$Kd_{bt} = \frac{₹1,20,000 + \frac{1}{5} (₹95,000)}{\frac{1}{2} (₹23,05,000)}$$

$$Kd_{bt} = \frac{₹1,39,000}{₹11,52,500}$$

$$Kd_{bt} = 0.1206 = \mathbf{12.06\%}$$

⇒ **After Tax**

$$Kd_{at} = \frac{\text{Interest}(1-t) + \frac{1}{n} (\text{Redeemable Value} - \text{Net Proceeds})}{\frac{1}{2} (\text{Redeemable Value} + \text{Net Proceeds})}$$

$$Kd_{at} = \frac{₹1,20,000 (1-0.5) + \frac{1}{5} (₹12,00,000 - ₹11,05,000)}{\frac{1}{2} (₹12,00,000 + ₹11,05,000)}$$

$$Kd_{at} = \frac{\text{₹}60,000 + \frac{1}{5}(\text{₹}95,000)}{\frac{1}{2}(\text{₹}23,05,000)}$$

$$Kd_{at} = \frac{\text{₹}79,000}{\text{₹}11,52,500}$$

$$Kd_{at} = 0.0685 = 6.85\%$$

Zero Coupon Bonds

Zero coupon bonds are bonds or debentures are issued at discount and redeemed at its face value or par value. No interest is payable on these types of debentures or bonds.

$$Kd_{\text{zero coupon bonds}} = LDF + \left(\frac{PVLDF}{PVLDF + PVHDF} \right) (HDF - LDF)$$

LDF = Lower discount factor

HDF = Higher discount factor

PVLDF = Present value of lower discount factor

PVHDF = Present value of higher discount factor

3. XYZ Ltd has issued redeemable zero-coupon bonds of ₹100 each at a discount rate of ₹60 repayable at the end of 4th year. Calculate the cost of zero coupon bond or debt.

Calculation of Zero Coupon Bond

Year	Cash flows	PV @13%	PVCF	PV @14%	PVCF
0	60	1	60	1	60
4	100	0.613	61.3	0.592	59.2
			1.3		(0.8)

$$Kd_{\text{zero coupon bonds}} = LDF + \left(\frac{PV \text{ LDF CF}}{PVLDFCF + PVHDFCF} \right) (HDF - LDF)$$

$$Kd_{\text{zero coupon bonds}} = 13\% + \left(\frac{1.3}{1.3 + 0.8} \right) (14 - 13)$$

$$Kd_{\text{zero coupon bonds}} = 13\% + (0.619)(14 - 13)$$

$$Kd_{\text{zero coupon bonds}} = 13\% + 0.619$$

$$Kd_{\text{zero coupon bonds}} = 13.619\%$$

PREFERENCE SHARE**➤ Irredeemable preference shares**

Irredeemable preference shares are those shares which cannot be redeemed. In other words, it will exist in the company for a life time.

$$K_p = \left[\frac{\text{Dividend}}{\text{Net Proceeds}} \right]$$

➤ Redeemable Preference Shares

Redeemable preference shares are a type of preference share, a company issues them to shareholders and later redeems them. This means the company can buy back the shares at a later date.

$$K_p = \frac{\text{Dividend} + \frac{1}{n} (RV - NP)}{\frac{1}{2} [RV + NP]}$$

4. A company issues 10000 at 10% preference shares of 100 each. Cost of issue is ₹2 per share. Calculate cost of preference shares if shares are issued
- a) At par
 - b) At premium of 10%
 - c) At discount of 5%

$$K_p = \left[\frac{\text{Dividend}}{\text{Net Proceeds}} \right]$$

(a) Dividend = ₹100*10% = ₹10

NP = FV-Issue Cost = ₹100- ₹2 = ₹98

$$K_p = \left[\frac{₹10}{₹98} \right]$$

$$K_p = 0.1020 = 10.20\%$$

(b) Dividend = ₹100*10% = ₹10

NP = FV + Premium-Issue Cost

= ₹100+ (₹100*10%)-₹2 = ₹108

$$K_p = \left[\frac{₹10}{₹108} \right]$$

$$K_p = 0.0926 = 9.26\%$$

$$(c) \text{ Dividend} = ₹100 * 10\% = ₹10$$

$$NP = FV - \text{Discount} - \text{Issue Cost}$$

$$= ₹100 - (₹100 * 5\%) - ₹2 = ₹93$$

$$K_p = \left[\frac{₹10}{₹93} \right]$$

$$K_p = 0.1075 = 10.75 \%$$

5. A company issue 8000, 10% preference shares of ₹100 each. Its redeemable after 10 years at a premium of 5%. The cost of issue is ₹ 2 per share. Calculate cost of preference shares.

$$K_p = \frac{\text{Dividend} + \frac{1}{n} (RV - NP)}{\frac{1}{2} (RV + NP)}$$

$$D = ₹100 * 10\% = ₹10$$

$$RV = (₹100 + (₹100 * 5\%)) = ₹105$$

$$NP = FV - \text{Issue Cost} = ₹100 - ₹2 = ₹98$$

$$n = 10 \text{ years}$$

$$K_p = \frac{₹10 + \frac{1}{10} (₹105 - ₹98)}{\frac{1}{2} (₹105 + ₹98)}$$

$$K_p = \frac{₹10 + 0.7}{\frac{1}{2} (₹203)}$$

$$K_p = \frac{₹10.7}{₹101.5}$$

$$K_p = 0.1054 = 10.54\%$$

COST OF EQUITY (K_e)**Methods**

- I. Dividend yield method
- II. Dividend yield plus growth rate method
- III. Capital Asset Pricing Model (**CAPM Model**)

Dividend yield method

$$K_e = \left[\frac{\text{Dividend}}{\text{Net Proceeds (or) Market Price}} \right]$$

6. A company issue 1500 equity shares of 100 each at a premium of 10%. The company has been 20% dividend to equity shareholders for the past 5 years & expects to maintain same in the future also. Compute the cost of equity. Will it make any difference if the market price of equity share is ₹160?

$$K_e = \left[\frac{\text{Dividend}}{\text{Net Proceeds (or) Market Price}} \right]$$

$$K_e = \left[\frac{₹100 * 20\%}{₹100 + (₹100 * 10\%)} \right]$$

$$K_e = \left[\frac{₹20}{₹110} \right]$$

$$K_e = .01818 = 1.818\%$$

⇒ Cost of equity when market price is ₹160

$$K_e = \left[\frac{\text{Dividend}}{\text{Net Proceeds (or) Market Price}} \right]$$

$$K_e = \left[\frac{₹20}{₹160} \right]$$

$$K_e = 0.125 = 12.5\%$$

Dividend yield plus growth rate method

⇒ **If dividend is declared at the beginning of the year**

$$K_e = \left[\frac{D_0 (1+g)}{\text{Net Proceeds (or) Market Price}} \right] + g$$

⇒ **If dividend is declared at the end of the year**

$$K_e = \left[\frac{D_1}{\text{Net Proceeds (or) Market Price}} \right] + g$$

D_0 = Dividend paid at beginning of the year

D_1 = Dividend paid at end of the year

g = Growth rate

NP = Net Proceed

MP = Market Price

7. The shares of a company are selling at ₹40 per share & it had paid dividend of ₹ 4/share in the last year. The investor expects company will grow in paying dividend of 5% per year.

- i. Compute the company cost of equity capital
- ii. If they anticipated the growth rate is 7% p.a then calculate the indicated market per share.

(i) cost of equity capital

$$K_e = \left[\frac{D_0 (1+g)}{\text{Net Proceeds (or) Market Price}} \right] + g$$

$$K_e = \left[\frac{₹4(1+0.05)}{₹40} \right] + 0.05$$

$$K_e = 0.105 + 0.05$$

$$K_e = 0.155 = 15.5\%$$

(ii) If they anticipated the growth rate is 7% p.a then the indicated market per share is

$$K_e = \left[\frac{D_0 (1+g)}{\text{Net Proceeds (or) Market Price}} \right] + g$$

$$0.155 = \left[\frac{₹4 (1+0.07)}{\text{Market Price}} \right] + 0.07$$

$$0.155 - 0.07 = \left[\frac{₹4.28}{\text{Market Price}} \right]$$

$$0.085 = \left[\frac{₹4.28}{\text{Market Price}} \right]$$

$$0.085 \text{ Market Price} = ₹4.28$$

$$\text{Market Price} = \frac{₹4.28}{0.085}$$

$$\text{Market Price} = ₹50.3529$$

Capital Asset Pricing Model (CAPM Model)

It explains behavior of share price or security price & provides a mechanism whereby investor would assist the impact of proposed securities investment on their overall portfolio risk & return. In other words, it formally describes the risk & return trade off securities.

$$K_e = R_f + \beta (R_m - R_f)$$

R_f = Risk Free Security

β = Beta variance

R_m = Market Return

$R_m - R_f$ = Risk Premium

Assumption of CAPM

1. All investor has same expectations regarding return & variance
2. All investors have same information on share price
3. There are no restrictions on investment
4. No corporate tax
5. No transaction cost
6. No single investor can influence on market price.

$$\begin{aligned} K_e &= R_f + \beta (R_m - R_f) \\ &= 0.08 + 1.2 (0.10 - 0.08) \\ &= 0.104 = 10.4\% \end{aligned}$$

8. Consider the following information to risk free rate of securities & market return of securities of A Ltd., during last 6 years

Year	Rf	Rm	Rj (security return)
1	0.06	0.14	0.08
2	0.05	0.03	0.11
3	0.07	0.21	0.29
4	0.08	0.26	0.25
5	0.09	0.03	0.107
6	0.07	0.11	0.104

On the basis of the above information you are required to determine cost of equity capital using CAPM approach.

$$K_e = R_f + \beta (R_m - R_f)$$

$$\beta = \frac{(\sum MJ - N\bar{M}\bar{J})}{(\sum M^2 - N\bar{M}^2)}$$

Year	Rf	Rm	M = Rm-Rf	M ²	Rj	J = Rj-Rf	MJ
1	0.06	0.14	0.08	0.0064	0.08	0.02	0.0016
2	0.05	0.03	(0.02)	0.0004	0.11	0.06	(0.0012)
3	0.07	0.21	0.14	0.0196	0.29	0.22	0.0308
4	0.08	0.26	0.18	0.0324	0.25	0.17	0.0306
5	0.09	0.03	(0.06)	0.0036	0.107	0.017	(0.00102)
6	0.07	0.11	0.04	0.0016	0.104	0.034	0.00136
	0.42	0.78	0.36	0.064		0.521	0.06214

$$\beta = \frac{(\sum MJ - N\bar{M}\bar{J})}{(\sum M^2 - N\bar{M}^2)}$$

$$\beta = \frac{(0.06214 - (6 \times 0.06 \times 0.0868))}{(0.064 - 6(0.06)^2)}$$

$$\beta = \frac{(0.06214 - 0.031248)}{(0.064 - 0.0216)}$$

$$\beta = \frac{0.030892}{0.0424}$$

$$\beta = 0.7286$$

$$\bar{M} = \frac{\sum M}{N} = \frac{0.36}{6} = 0.06$$

$$\bar{J} = \frac{\sum J}{N} = \frac{0.521}{6} = 0.0868$$

Cost of Equity using CAPM

$$K_e = R_f + \beta (R_m - R_f)$$

$$K_e = \frac{0.42}{6} + 0.7286 \left\{ \frac{0.78}{6} - \frac{0.42}{6} \right\}$$

$$K_e = 0.07 + 0.7286 \{0.13 - 0.07\}$$

$$K_e = 0.07 + 0.0437$$

$$K_e = 0.1137 = 11.37\%$$

COST of CAPITAL (K_o)**OR****Weighted Average Cost of Capital (WACC)**

Also called as overall cost of capital or composite cost of capital. The weighted average cost of capital is calculated by multiplying the weights of the various sources of capital with the cost of capital of that particular source. Weight is the proportion of each source of fund in the capital structure.

Steps involved in calculation of Weighted Average Cost of Capital

1. Calculate the cost of specific sources of funds
2. Multiply the cost of specific sources by its proportion in the capital structure (weight)
3. Add the weighted component costs to get the firm's weighted average cost of capital

$$\text{WACC (OR) } K_o = \text{Proportion of capital} * \text{After tax cost}$$

OR

$$\text{WACC (OR) } K_o = \left\{ \frac{\text{Total Weighted Cost}}{\text{Total Sources of funds}} \right\} * 100$$

1. A firm has the following capital structure & after-tax costs for the different sources of funds used

Sources of funds	Amount (₹)	Proportion (%)	After tax cost
Debt	15,00,000	25	5
Preference shares	12,00,000	20	10
Equity shares	18,00,000	30	12
Retained earnings	15,00,000	25	11
	60,00,000	100	

You are required to compute the weighted average cost of capital

Solution: -

Calculation of Weighted Average Cost of Capital

$$\text{WACC (OR) } K_o = \text{Proportion of capital} * \text{After tax cost}$$

Sources of funds	Amount (₹)	Proportion	After tax cost	Cost of Weighted
Debt	15,00,000	0.25	0.05	0.0125
Preference shares	12,00,000	0.2	0.1	0.02
Equity shares	18,00,000	0.3	0.12	0.036
Retained earnings	15,00,000	0.25	0.11	0.0275
Total Weighted Average Cost of capital				0.096

Total Weighted Average Cost of capital = 0.096=9.6%

MODULE – 6**WORKING CAPITAL MANAGEMENT / NET CURRENT ASSET MANAGEMENT**

Working Capital is the amount of funds necessary to cover the cost of operating the enterprise. Working Capital Management (**WCM**) is the process of managing current assets and Current liabilities to ensure the short-term liquidity of firm.

Components of Current Assets

- Cash in hand
- Cash at bank
- Debtors
- Stock
- Bills Receivables
- Prepaid Expenses
- Outstanding income
- Short-term investment

Components of Current Liabilities

- Creditors
- Bills Payable
- Bank Overdraft
- Prepaid income
- Outstanding expenses
- Short-term loans

Types of Working Capital

1. Gross Working Capital = Total current assets
2. Net Working Capital = TCA – TCL
3. Permanent **or** fixed working capital
4. Variable **or** fluctuating working capital
5. Positive working capital = TCA more than TCL
6. Negative working capital = TCA is less than TCL

Factors Influencing on Working Capital (or) net current asset

1. Nature of the business
2. Size of the business
3. Business cycle (Or) Production cycle
4. Profit level
5. Dividend policy
6. Terms of purchases & sales **or** credit policy
7. Operating cycle
8. Growth & expansion activity
9. Availability of raw materials
10. Seasonal variation in demand & supply
11. Price level changes
12. Level of taxes

Approaches for financing working capital management

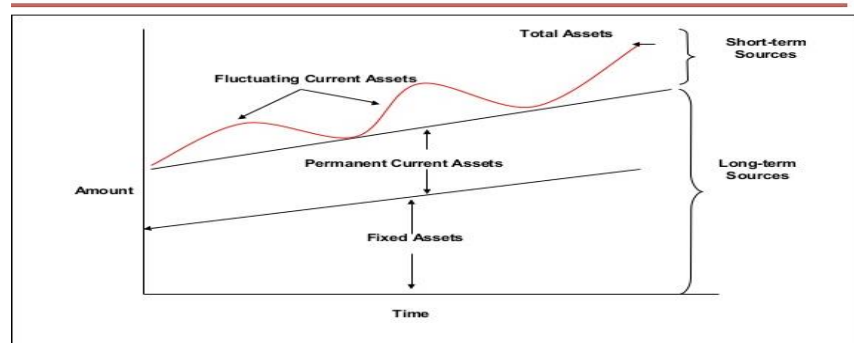
There are basically three **approaches to financing working capital**. These are:

1. Matching Approach (Or) Hedging Approach

The hedging approach is also known as the matching approach. Under this approach, the funds for acquiring fixed assets and permanent current should be acquired with long term funds and for temporary current assets short term funds should be used.

$$\text{LTS} = \text{FA} + \text{PCA}$$

$$\text{STS} = \text{Fluctuating CA}$$

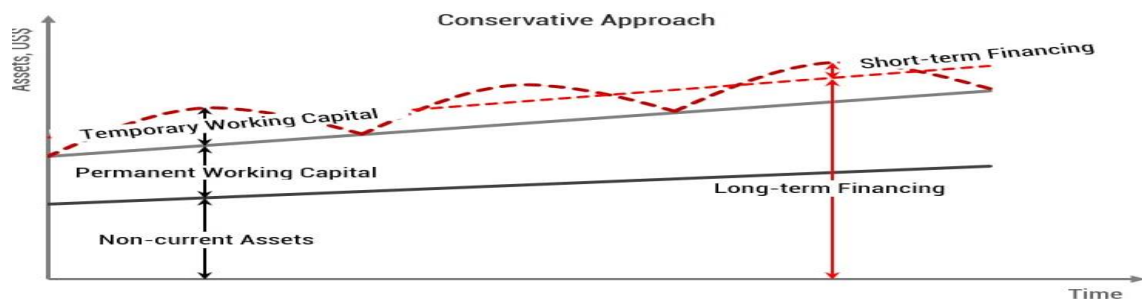
Hedging approach to assets financing

2. Conservative Approach:

This approach suggests that in addition to fixed assets and permanent current assets, even a part of variable current assets should be financed from long-term sources. The short-term sources are used only to meet the peak seasonal requirements.

$$\text{LTS} = \text{FA} + \text{PCA} + \text{Portion of Fluctuating CA}$$

$$\text{STS} = \text{Portion of Fluctuating CA}$$

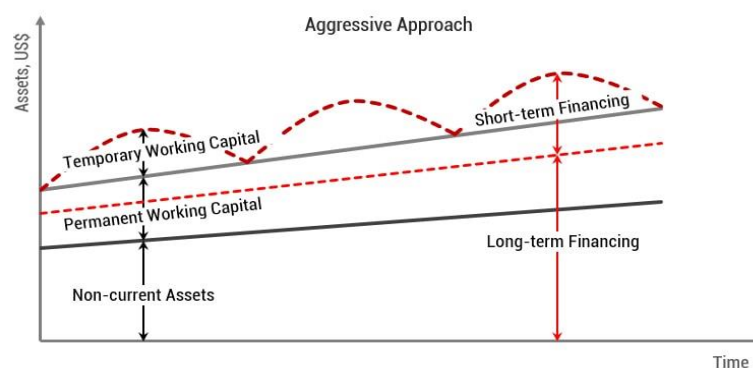


3. Aggressive Approach:

This approach depends more on short-term funds. More short-term funds are used particularly for variable current assets and a part of even permanent current assets; the funds are raised from short term sources. Under this approach current assets are maintained just to meet the current liabilities without keeping cushions for the variation in working capital needs.

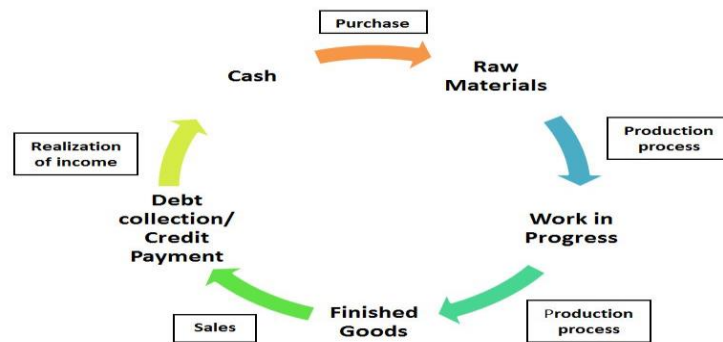
$$\text{LTS} = \text{FA} + \text{Portion of PCA}$$

$$\text{STS} = \text{Portion of PCA} + \text{Fluctuating CA}$$



Operating Cycle & Cash Cycle

Operating Cycle is time required for converting cash to raw-material, raw-material to work-in-progress, work-in-progress to finished goods, finished goods to debtors, debtors to cash.



Operating Cycle of a Business

Cash Cycle is time required for converting cash to raw-material, raw-material to work-in-progress, work-in-progress to finished goods, and finished goods to cash sales.



Formulas for Operating Cycle & Cash Cycle

Operating Cycle = Raw material Conversion Period + WIP Conversion Period +
Finished goods Conversion Period + Debtors Conversion Period

Cash Cycle = Operating Cycle – Creditors Conversion Period

⇒ **Raw material Conversion Period**

$$= \frac{\text{Stock of raw material}}{\text{raw material Consumption per day}}$$

⇒ **WIP Conversion Period**

$$= \frac{\text{Stock of WIP}}{\text{Cost of production per day}}$$

⇒ **Finished Goods Conversion Period**

$$= \frac{\text{Stock of finished goods}}{\text{Cost of goods sold per day}}$$

⇒ **Debtors (or) Account Receivables Conversion Period**

$$= \frac{\text{Debtors}}{\text{Net credit sales per day}}$$

⇒ **Creditors (or) Account Payable Conversion Period**

$$= \frac{\text{Creditors}}{\text{Net credit purchases per day}}$$

Problems on operating cycle & cash cycle

1. From the following information extracted from the books of a manufacturing concern, compute operating cycle & cash cycle (Net operating cycle)

Period covered	365 days
Average period of credit allowed by suppliers	16 days
Avg. of total debtors outstanding (₹)	480
Raw material consumption (₹)	4400
Total production cost (₹)	10000
Total cost of goods sold for the year (₹)	10500
Sales for the year (₹)	16000
Value of avg, stock maintained	
Raw – material	320
Work in progress	350
Finished goods	260

Solution**⇒ Raw material Conversion Period**

$$\begin{aligned}
 &= \frac{\text{Stock of raw material}}{\text{Raw material consumption per day}} \\
 &= \frac{320}{\left(\frac{4400}{365}\right)} \\
 &= 26.56 = 27 \text{ days}
 \end{aligned}$$

⇒ WIP Conversion Period

$$\begin{aligned}
 &= \frac{\text{Stock of WIP}}{\text{Cost of production per day}} \\
 &= \frac{350}{\left(\frac{10000}{365}\right)} \\
 &= 12.77 = 13 \text{ days}
 \end{aligned}$$

⇒ Finished Goods Conversion Period

$$\begin{aligned}
 &= \frac{\text{Stock of finished goods}}{\text{Cost of goods sold per day}} \\
 &= \frac{260}{\left(\frac{10500}{365}\right)} \\
 &= 9.03 = 9 \text{ days}
 \end{aligned}$$

⇒ **Debtors (or) Account Receivables Conversion Period**

$$\begin{aligned} &= \frac{\text{Debtors}}{\text{Net credit sales per day}} \\ &= \frac{480}{\left(\frac{16000}{365}\right)} \\ &= 10.95 = \mathbf{11 \text{ days}} \end{aligned}$$

⇒ **Creditors (or) Account Payable Conversion Period**

$$\begin{aligned} &= \frac{\text{Creditors}}{\text{Net credit purchases per day}} \\ &= \mathbf{16 \text{ days}} \end{aligned}$$

Operating Cycle = Raw material Conversion Period + WIP Conversion Period +
Finished goods Conversion Period + Debtors Conversion Period
= 27 days + 13 days + 9 days + 11 days
= **60 days**

Cash Cycle = **Operating Cycle** – **Creditors Conversion Period**
= 60 days – 16 days
= **44 days**

Format for the estimation of Working capital

SL No.	Particulars	Amount	
I.	CURRENT ASSEST		
i	Raw Material		*****
ii	Work in Progress (WIP)		
	Raw Material	***	
	Wages / Labour overhead	***	
	Other overheads	***	*****
iii	Finished goods		*****
iv	Sundry debtors		*****
v	Cash & bank Balance		*****
vi	Any payments in advance		*****
Total Current Assets (I)			*****
II	CURRENT LIABILITIES		
i	Sundry creditors		*****
ii	Lag or delay in payments		
	Wages	***	
	Overheads	***	
	Selling & distribution expenses	***	
	Office & administrative expenses	***	
iii	Any prepaid income		*****
Total Current Liabilities (II)			*****
Net Working Capital (I –II)			*****
Add: - Contingency (or) Provision			*****
Required Net working Capital			*****

Note: -

- Direct wages or labor & overheads are called conversion cost. Depending upon information given in problem should be considered for work done for the calculation of WIP. In absence of specific information, it should be assumed that 50% of work done for WIP calculations of wages & overheads.
- In case of finished goods & debtors cost of raw materials, wages & overheads are fully considered for the calculation propose.

Formula for the calculation all current assets & liabilities

$$= \text{Number of budgeted units} * \text{Cost per unit} * \text{Holding period}$$

Problems on Working capital

1. While preparing a project report on behalf of a client you have collected the following facts. Estimate the net working capital required for that project. Add 10% to you computed figures to allow contingencies.

Particulars	Amount per unit
Estimated cost per unit of production	
Raw material	₹80
Direct Labour	₹30
Overheads (exclusive depreciation, ₹ 10 per unit)	₹60
Total cost	₹170

Additional Information

- Selling price ₹ 200 per unit
- Level of activity 1,04,000 units of production per annum
- Raw material in stock, average 4 weeks
- Work in Progress (assume 50% completion stage in respect of conversion costs & 100 completion in respect of materials), average 2 weeks
- Finished goods in stocks, average 4 weeks
- Credit allowed by suppliers, average 4 weeks
- Credit allowed to debtors, average 8 weeks
- Lag in payment of wages, average 1.5 weeks
- Cash at bank is expected to be ₹ 25,000

You may assume that production is carried on evenly throughout the year (52 weeks) & wages & overheads accrue similarly. All sales are on credit basis only.

Solution**Estimation of required working capital for a project**

SL No.	Particulars	Amount	
I.	CURRENT ASSEST		
i	Raw-Material $\{104000 \text{ units} * ₹80 * \frac{4}{52}\}$		6,40,000
ii	Work-in Progress		
	Raw material $\{104000 \text{ units} * ₹80 * \frac{2}{52}\} * 100\%$	3,20,000	
	Wages $\{104000 \text{ units} * ₹30 * \frac{2}{52}\} * 50\%$	60,000	
	Overhead $\{104000 \text{ units} * ₹60 * \frac{2}{52}\} * 50\%$	1,20,000	5,00,000
iii	Finished goods $\{104000 \text{ units} * ₹170 * \frac{4}{52}\}$		13,60,000
iv	Sundry Debtors $\{104000 \text{ units} * ₹170 * \frac{4}{52}\}$		27,20,000
v	Cash at bank		25,000
	Total Current Assets (I)		52,45,000
II	CURRENT LIABILITIES		
i	Sundry Creditors $\{104000 \text{ units} * ₹80 * \frac{4}{52}\}$		6,40,000
ii	Lag in payment of wages $\{104000 \text{ units} * ₹30 * \frac{1.5}{52}\}$		90,000
	Total Current Liabilities (II)		7,30,000
	Net Working Capital (I – II)		45,15,000
Add:- Contingency (or) Provision (₹45,15,000*10%)			4,51,500
	Required Net working Capital		49,66,500

2. XYZ Ltd sells goods at a gross profit of 20%. It includes depreciation as a part of cost of production. The following figures for the 12 months – period ending March 31st, current year is given you to ascertain the requirements of working capital of the company on a cash basis

In your working, you are required to assume that

- (i) A safety margin of 15% will be maintained
 - (ii) Cash is to be held to the extent of 50% of current liability
 - (iii) There will be no work in progress
 - (iv) Tax is to be ignored
 - (v) Finished goods are to be valued at manufacturing costs. Stocks of raw materials & finished goods are kept at one month's requirement
- Sales at 2 month's credits. ₹27,00,000
 - Materials consumed (suppliers credit is for 2 months), ₹6,75,000
 - Wages (paid on the last day of the month) ₹5,40,000

- Manufacturing expenses outstanding at the end of the year (cash expenses are paid one month in arrears), ₹60,000
- Total administrative expenses (paid as above), ₹1,80,000
- Sales promotion expenses (paid quarterly in advance) ₹90,000

Solution
Estimation of working capital for XYZ Ltd.,

SL No.	Particulars	Amount	
I.	CURRENT ASSETS		
i	Raw material $\{₹6,75,000 * \frac{1}{12}\}$		56,250
ii	Work-in progress		----
iii	Finished goods (RM+Wages+Manuf.Expen)		
	$\{ (₹6,75,000 + ₹5,40,000 + ₹60,000) * \frac{1}{12} \}$		1,06,250
iv	Sundry debtors $\{ (₹27,00,000 - 20\%) * \frac{2}{12} \}$		3,60,000
v	Cash balance $\{ ₹1,55,000 * 50\% \}$		77,500
vi	Sales promotion expenses paid in advance $\{ ₹90,000 * \frac{3}{12} \}$		22,500
Total Current Assets (I)			6,22,500
II	CURRENT LIABILITIES		
i	Sundry creditors $\{ ₹6,75,000 * \frac{2}{12} \}$		1,12,500
ii	Lag or delay or arrears in payments		
	Wages $\{ ₹5,40,000 * \frac{0.5}{12} \}$	22,500	
	Manufacturing expenses $\{ ₹60,000 * \frac{1}{12} \}$	5,000	
	Administrative Expenses $\{ ₹1,80,000 * \frac{1}{12} \}$	15,000	42,500
Total Current Liabilities (II)			1,55,000
Net Working Capital (I – II)			4,67,500
Add: - 15% Contingency (or) Provision $(₹4,67,500 * 15\%)$			70,125
Required Net working Capital			5,37,625

MODULE – 5**INVESTMENT DECISION {CAPITAL BUDGETING}**

Long term Investment Decisions (Capital Budgeting): Need and importance of capital budgeting and its process-Techniques of capital budgeting – (Payback period, accounting rate of return, Net present value, Internal rate of return, Profitability index method,). (Case Study) (Theory & Problem).

Capital budgeting or investment decision is a process of evaluating & selecting long term investment or expenditure that are consistent with the goal of shareholders wealth.

Process/Steps in Capital Budgeting

1. Identification of investment proposals
2. Screening the various proposals
3. Evaluating various proposals
4. Fixing priorities
5. Final approval & preparation of capital budget
6. Implementing proposal
7. Proposal review

Techniques /Methods of Capital Budgeting**I. Non-Discounting Technique**

1. Pay Back Period (PBP)
 - ➔ Discounted Pay Back Period (DPBP)
2. Accounting Rate of Return/Average Rate of Return (ARR)

II. Discounting Technique

1. Net Present Value (NPV)
 - ➔ Modified Net Present Value (Modified NPV)
2. Internal Rate of Return (IRR)
 - ➔ Modified Internal Rate of Return (MIRR)
3. Profitability Index (PI) or Benefit Cost Ratio (BCR)

Format for the calculation of Cash flow

Earnings Before Interest, Depreciation & Tax (EBID&T)	****
Less: - Interest	***
Earnings Before Depreciation & Tax (EBDT)	****
Less: - Depreciation	***
Earnings After Depreciation & Before Tax (EAD&BT)	*****
Less: - Tax	***
Earnings After Depreciation & Tax (EAD&T)	****
Add: - Depreciation	***
Earnings After Tax & Before Depreciation (EAT&BD)	****

I. Pay Back Period (PBP)

Pay Back Period (PBP) represents the time period required for complete recovery of the initial investment in the project.

It is the period within which the total cash inflows from the project equals the cost of investment in the project, lower the pay back period better the project.

⇒ Advantages of Payback back Period

1. It is very simple and easy to understand.
2. It is one of the oldest techniques
3. There is no need of specialized knowledge

⇒ Disadvantages of Payback back Period

1. It does not consider cash flows after payback.
2. Does not consider time value of money
3. Does not consider the element of risk

→ When Cash Flows are Equal

$$\text{Pay Back Period} = \frac{\text{Initial Investment (or) Cash Outlay}}{\text{Annual Cash Inflow}}$$

→ When Cash Flows are Unequal (OR) Varies

$$\text{PBP} = \text{Year in which cash inflow is nearer to cash outflow} + \frac{\text{Amount yet to recover}}{\text{Cash inflow of next year}}$$

Note: - Cash flow should be after tax & before depreciation

Problems

1. A project costs ₹ 1, 00,000 & yield annual cash inflows of ₹ 20000 for 8 years.
Calculate its pay -back period?

Solution: -

$$\begin{aligned}\text{Pay Back Period} &= \frac{\text{Initial Investment (or) Cash Outlay}}{\text{Annual Cash Inflow}} \\ \text{Pay Back Period} &= \frac{₹1,00,000}{₹20,000} = \mathbf{5 \text{ years}}\end{aligned}$$

2. Determine pay - back period for a project which requires a cash outlay of ₹10,000 & generates cash inflows of ₹2000, ₹4000, ₹3000 & ₹2000 in the 1st, 2nd, 3rd & 4th year respectively.

Solution: -

Year	Cash Inflow	Cummulative Cash Inflow
1	2,000	2,000
2	4,000	6,000
3	3,000	9,000
4	2,000	11,000

$$\text{PBP} = \text{Year in which cash inflow is nearer to cash outflow} + \frac{\text{Amount yet to recover}}{\text{Cash inflow of next year}}$$

$$\text{PBP} = 3 + \frac{₹10,000 - ₹9,000}{₹2,000}$$

$$\mathbf{\text{PBP}=3.5 \text{ years}}$$

II. Accounting Rate of Return /Average Rate of Return (ARR)**Accounting Rate of Return**

$$\text{Accounting Rate of Return} = \left\{ \frac{\text{Average EAT \& AD}}{\text{Initial Investment} + \text{Working Capital}} \right\} * 100$$

Average Rate of Return /ARR

$$\text{Avg. Rate of Return} = \left\{ \frac{\text{Average EAT \& AD}}{\left(\frac{\text{Initial Investment} - \text{Scrap Value}}{2} \right) + \text{WC} + \text{SV}} \right\} * 100$$

3. From the following determine the **accounting rate of return**

Particulars	Machine - A	Machine -B
Original cost	56,125	56,125
Working capital	5,000	6,000
Estimated life (years)	5	5
Salvage value	3000	3000
Tax rate	55%	55%
years	Annual estimated income after depreciation & taxes	
1	3375	11375
2	5375	5375
3	7375	7375
4	9375	5375
5	11375	3375

Solution: -

$$\text{Accounting Rate of Return} = \left\{ \frac{\text{Average EAT \& AD}}{\text{Initial Investment} + \text{Working Capital}} \right\} * 100$$

Machine 'A'

Accounting Rate of Return

$$= \left\{ \frac{(\text{₹}3,375 + \text{₹}5,375 + \text{₹}7,375 + \text{₹}9,375 + \text{₹}11,375)/5}{\text{₹}56,125 + \text{₹}5,000} \right\} * 100$$

$$\text{Accounting Rate of Return} = \left\{ \frac{\text{₹}36,875/5}{\text{₹}61,125} \right\} * 100$$

$$\text{Accounting Rate of Return} = 0.1206 * 100 = 12.06\%$$

III. Net Present Value (NPV)

The net present value method is a modern method of evaluating investment proposal. This method takes into consideration the time value of money.

NPV can be defined as difference between present value of cash inflows and present value of cash outflows of the project.

⇒ **Advantages of NPV**

1. It takes into account the time value of money.
2. It takes into consideration all cash flows occurring over the entire life of the project including scrap value.
3. It is useful for selection of mutually exclusive projects.
4. It takes into consideration the objective of maximum profitability.

⇒ **Disadvantages of NPV**

1. It is difficult to understand and calculate.
2. It is not easy to determine an appropriate discount rate.
3. It may not be accurate to compare projects with uneven life

Accept & Reject Criteria

➔ **Accept – When NPV is Positive**

➔ **Reject – When NPV is Negative**

Note: - Cash flow should be after tax & before depreciation

NPV= Total Present Value of Cash inflow after tax & before depreciation

—

Initial Investment

NPV Total PV of CFAT&BD -Initial Investment

IV. Benefit Cost Ratio (BCR) / Profitability Index (PI)

Benefit cost ratio / Profitability Index refers to the ratio of the present value of all cash inflows to the present value of all cash outflows of the project.

PI can be interpreted as the amount of cash inflow available per rupee of cash outflow.

$$\text{BCR (Or) PI} = \frac{\text{Total PV of Cash flow after tax \& before depreciation}}{\text{Initial Investment}}$$

Cash flow should be after tax but before depreciation

Accept & Reject Criteria

Accept = PI is more than 1

Reject = PI is less than 1

Advantages of BCR/PI

- It considers time value of money
- It considers cash inflow throughout the life of the project
- It attempts to find out the maximum rate of interest at which funds are invested in project

Disadvantages of BCR/PI

- It may give inconsistent results with NPV methods
- It difficult to understand

4. From the following information calculate the net present value of the two projects & suggest which of the projects should be accepted assuming a discount rate of 10%.

Particulars	Project – X	Project - Y
Initial investment	20,000	30,000
Estimated life (years)	5	5
Scrap value	1,000	2,000
Profit before depreciation & after taxes are as follows		
1 (years)	5,000	20,000
2	10,000	10,000
3	10,000	5,000
4	3,000	3,000
5	2,000	2,000

Solution: -

⇒ **Project ‘X’**

(i) **Calculation of Net Present Value**

Year	CFAT&BD	PV @10%	PV of CFAT&BD
1	5,000	0.909	4,545
2	10,000	0.826	8,260
3	10,000	0.751	7,510
4	3,000	0.683	2,049
5	2,000	0.621	1,242
Total PV of CFAT&BD			23,606
Less:- Initial Investment			20,000
NPV			3,606

(ii) **Benefit Cost Ratio /Profitability Index**

$$\text{BCR (Or)PI} = \frac{\text{Total PV of Cash flow after tax \& before depreciation}}{\text{Initial Investment}}$$

$$\text{BCR (Or)PI} = \frac{\text{₹23,606}}{\text{₹20,000}} = 1.1803$$

⇒ **Project ‘Y’**

$$\text{NPV} = \text{₹3,486}$$

$$\text{BCR/PI} = 1.1162$$

Recommendation: - Company better to accept project 'X' because its gives more positive NPV & BCR compare project 'Y'

5. A company has 2 project options which are exclusive for increasing the plant's capacity. The management of the company has estimated under 3 conditions; optimistic, most likely & pessimistic.

Particulars	Project –A	Project -B
Investment cost for the project	2,00,000	2,00,000
<u>Cash inflows</u>		
Optimistic	50,000	35,000
Most likely	30,000	28,000
Pessimistic	14,000	15,000

- Calculate the NPV with each estimate for the projects. The size of both projects is 15 years & cost of capital is 12%.
- Which project you would select? Show your calculation. Present value factor @ 12% for 15 years is 6.811.

Solution: -

(i) Calculation of NPV

⇒ **Project 'A'**

Situation	Cash Flow	PV @12%	PV of cash flow	NPV
Optimistic	50,000	6.811	3,40,550	1,40,550
Most likely	30,000	6.811	2,04,330	4,330
Pessimistic	14,000	6.811	95,354	(1,04,646)
Overall NPV				40,234

⇒ **Project 'B'**

Situation	Cash Flow	PV @12%	PV of cash flow	NPV
Optimistic	35,000	6.811	2,38,385	38,385
Most likely	28,000	6.811	1,90,708	(9,292)
Pessimistic	15,000	6.811	1,02,165	(97,835)
Overall NPV				(68,742)

- Company better to accept project "A". because it gives overall positive NPV compare to project 'B'

Internal Rate of Return (IRR)

IRR is the rate at which the present value of cash inflows will be equal to the present value of outflows, i.e., the rate at which NPV = Zero.

IRR is the rate of return that a project earns. It is defined as the discount rate which equates the aggregate present value of the net cash inflows (EBDAT) with the aggregate present value of cash outflows.

In other words, it is the rate which gives the project NPV of Zero.

Advantages of IRR

- It takes into time value of money.
- It considers cash flows throughout the projects life.
- It does not use the concept of required rate of return or the cost of capital; itself provide a rate of return which is indicative of profitability.

Disadvantages of IRR

- It is difficult to understand and calculate.
- It produces multiple rates of returns which can be confusing.
- IRR is not reliable when compared to NPV

$$\text{IRR} = \text{LDF} + \left\{ \frac{\text{PV of LDF} - \text{Initial Investment}}{\text{PV of LDF} - \text{PV of HDF}} \right\} * \{\text{HDF} - \text{LDF}\}$$

Accept & Reject Criteria

Accept = More than cost of capital

Reject = Less than cost of capital

Cash flows should be after tax but before depreciation

6. From the following information calculate IRR. The initial investment of ₹7,50,000 of proposal @ discount factor 10%.

Year	1	2	3	4	5	6
PATBD	1,65,000	1,40,000	2,15,000	1,88,000	1,07,000	80,000

Solution: -

Calculation of IRR

Year	PATBD	PV @10%	PV of PATBD	PV @7%	PV of PATBD
1	1,65,000	0.909	1,49,985	0.935	1,54,275
2	1,40,000	0.826	1,15,640	0.873	1,22,220
3	2,15,000	0.751	1,61,465	0.816	1,75,440
4	1,88,000	0.683	1,28,404	0.763	1,43,444
5	1,07,000	0.621	66,447	0.713	76,291
6	80,000	0.564	45,120	0.666	53,280
Total PV of PAT&BD			6,67,061		7,24,950
Less:- Initial Investment			7,50,000		7,50,000
NPV			(82,939)		(25,050)

Year	PATBD	PV @5%	PV of PATBD	PV @6%	PV of PATBD	
1	1,65,000	0.952	1,57,080	0.943	1,55,595	
2	1,40,000	0.907	1,26,980	0.890	1,24,600	
3	2,15,000	0.864	1,85,760	0.840	1,80,600	
4	1,88,000	0.823	1,54,724	0.792	1,48,896	
5	1,07,000	0.784	83,888	0.747	79,929	
6	80,000	0.746	59,680	0.705	56,400	
Total PV of PAT&BD			7,68,112		7,46,020	
Less:- Initial Investment			7,50,000		7,50,000	
NPV			18,112		(3,980)	

$$\text{IRR} = \text{LDF} + \left\{ \frac{\text{PV of LDF} - \text{Initial Investment}}{\text{PV of LDF} - \text{PV of HDF}} \right\} * \{\text{HDF} - \text{LDF}\}$$

$$\text{IRR} = 5 + \left\{ \frac{₹7,68,112 - ₹7,50,000}{₹7,68,112 - ₹7,46,020} \right\} * \{6 - 5\}$$

$$\text{IRR} = 5 + \left\{ \frac{₹18,112}{₹22,092} \right\} * \{1\}$$

$$\text{IRR} = 5.82\%$$

Modified IRR (MIRR)

Modified internal rate of return (MIRR) is a similar technique to IRR. Technically, MIRR is the IRR for a project with an identical level of investment and NPV to that being considered but with a single terminal payment.

Step1: - Calculation of Present Value of Cost (PVC)

$$\sum_{t=0}^n = \frac{\text{Cash Outflow}}{(1 + r)^t}$$

Step 2: - Calculation of Present Value of Terminal Value

$$\sum_{t=0}^n = \text{Cash Inflow } (1 + r)^{n-t}$$

Step 3: - Calculation of Modified IRR

$$(1 + \text{MIRR})^n = \frac{\text{Present Value of Terminal Value}}{\text{Present Value of Cost}}$$

7. 'P' Ltd is evaluating a project that has the following cash flow stream associated with it

Year	0	1	2	3	4	5	6
Cash flows (₹ in millions)	-120	-80	20	60	80	100	120

The cost of capital for P Ltd is 15%. Calculate the MIRR

Calculation of MIRR**Step1: - Calculation of Present Value of Cost (PVC)**

$$\sum_{t=0}^n = \frac{\text{Cash Outflow}}{(1 + r)^t}$$

$$\sum_{t=0}^1 = \frac{₹120 \text{ million}}{(1 + 0.15)^0} + \frac{₹80 \text{ million}}{(1 + 0.15)^1}$$

$$\sum_{t=0}^1 = ₹120 \text{ million} + ₹ 69.57 \text{ million}$$

$$\text{PVC} = ₹189.57 \text{ million}$$

Step 2: - Calculation of Present Value of Terminal Value

$$\sum_{t=0}^n = \text{Cash Inflow } (1 + r)^{n-t}$$

$$\begin{aligned} \text{PV of TV} = \sum_{t=2}^6 = & ₹20 \text{ M } (1+0.15)^{6-2} + ₹60 \text{ M } (1+0.15)^{6-3} + ₹80 \text{ M } (1+0.15)^{6-4} \\ & + ₹100 \text{ M } (1+0.15)^{6-5} + ₹120 \text{ M } (1+0.15)^{6-6} \end{aligned}$$

$$\text{PV of TV} = ₹ 34.98 \text{ M} + ₹91.25 \text{ M} + ₹ 105.8\text{M} + ₹115 \text{ M} + ₹120$$

$$\text{PV of TV} = ₹467.03 \text{ Million}$$

Step 3: - Calculation of Modified IRR

$$(1 + \text{MIRR})^n = \frac{\text{Present Value of Terminal Value}}{\text{Present Value of Cost}}$$

$$(1 + \text{MIRR})^6 = \frac{₹467.03 \text{ million}}{₹189.57 \text{ million}}$$

$$(1 + \text{MIRR})^6 = ₹2.4636 \text{ million}$$

$$1 + \text{MIRR} = (₹2.4636 \text{ million})^{\frac{1}{6}}$$

$$1 + \text{MIRR} = 1.16$$

$$\text{MIRR} = 1.16 - 1$$

$$\text{MIRR} = 0.16 = 16\%$$

MODULE – 6**CAPITAL STRUCTURE & DIVIDEND POLICY**

Capital structure and Dividend Decisions: Capital structure– Planning the capital structure- optimum capital structure- determination of capital structure- Leverages- EBIT and EPS analysis. (Theory & Problem).

Dividend decisions & policies – Factors affecting the dividend policy – types of Dividend Policy- forms of dividend-bonus issue-stock split (Theory only)

Capital Structure

Capital structure of a company refers to the composition or make-up of its capitalization and it include all long-term sources of capital such as equity shares, preference shares, debenture, bonds, loans etc.

Capital structure is a mixture or proportionate of different sources of capital i.e., equity & debt.

Optimum capital structure

It is a capital structure at which weighted average cost of capital or overall cost is minimum & maximizes the value of firm.

In other words, the combination of debt & equity that leads to minimizing overall cost of capital & maximizing the value of firm.

Assumptions of capital structure

1. There are only two sources of finance that are used in business i.e., equity & debt
2. Dividend payout ratio is 100%
3. Total assets are given & don't change or investment decision assumed to be constant
4. Total finance or capital remains constant but firm can change degree of leverage or proportionate
5. Operating profit (EBIT) are not expected to grow
6. All investors are assumed to have same subjective probability distribution of the future expected EBIT given time for a firm.
7. Business risk is constant over a period of time

8. Perpetual or infinite life of firm.

Factors influencing on capital structure

1. Growth & stability of sales
2. Cost of capital
3. Nature & size of the firm
4. Cash flow ability
5. Control
6. Flexibility
7. Requirement of investors
8. Capital market conditions
9. Asset structure
10. Purpose of finance
11. Period of finance
12. Corporate tax rate
13. Legal requirements

Leverages

It is the employment of assets or sources of finance for which from pay fixed cost or return. In financial management the term 'leverage' is used to describe the firm's ability to use fixed assets or funds to increase the returns of the owners; i.e., shareholder.

Types of Leverages

1. Operating leverage
2. Financial leverage
3. Combined/total leverage

Operating leverage

It is the measurement of the degree to which a firm or project incurs a combination of fixed & variable cost.

It is the measure of how revenue growth translates into growth in operating income.

$$\text{Operating Leverage} = \text{Contribution} / \text{Operating Profit (EBIT)}$$

Financial leverage

It is considered with the effect of changes of EBIT on earnings available to equity shareholders.

$$\text{Financial Leverage} = \text{Operating Profit (EBIT)} / \text{Earning Before Tax (EBT)}$$

Combined leverage

It is the combination of operating & financial leverage

$$\text{CL} = \text{Contribution} / \text{Earning Before Tax}$$

$$\text{CL} = \text{Operating leverage} * \text{Financial leverage}$$

Income Statement

Particulars	Amount
Sales	***
Less: - Variable Cost	***
Contribution	***
Less: - Fixed Cost	***
Operating Profit (EBIT)	***
Less: - Interest	***
Earnings Before Tax (EBT)	***
Less: - Tax	***
Earnings After Tax (EAT)	***
Less: - Preference Dividend	***
Earnings Available to equity shareholders	***

1. PQR Ltd. has sales of ₹40 Lakhs. Variable cost 70% of the sales & fixed cost is ₹8 lakhs. The firm has raised ₹ 20 lakhs funds by issue of debentures at the rate of 10%. Compute Operating Leverage, Financial Leverage & Combined Leverage.

Income statement of PQR Ltd

Particulars	Amount
Sales	40,00,000
Less: - Variable Cost (70% on sales)	28,00,000
Contribution	12,00,000
Less: - Fixed Cost	8,00,000
Operating Profit (EBIT)	4,00,000
Less: - Interest (₹20,00,000*10%)	2,00,000
Earnings Before Tax (EBT)	2,00,000

⇒ **Operating Leverage**

$$= \frac{\text{Contribution}}{\text{EBIT}} = \frac{₹12,00,000}{₹4,00,000} = 3 \text{ times}$$

⇒ **Financial Leverage**

$$= \frac{\text{Operating Profit (EBIT)}}{\text{EBT}} = \frac{₹4,00,000}{₹2,00,000} = 2 \text{ times}$$

⇒ **Combined Leverage**

$$= \frac{\text{Contribution}}{\text{EBT}} = \frac{₹12,00,000}{₹2,00,000} = 6 \text{ times}$$

OR

Combined Leverage=OL*FL

=3 times*2 times

= 6 times

2. PQR Ltd. has sales of ₹60 Lakhs. Variable cost 60% of the sales & fixed cost is ₹5 lakhs. The firm has raised ₹25 lakhs funds by issue of debt at the rate of 12%. Compute Operating Leverage, Financial Leverage & Combined Leverage.
3. A firm has sales of ₹10 lakh variable cost of ₹7 lakh & fixed cost of ₹2 lakh & debt of ₹5 lakh at 10% rate of interest. What are the operating, financial & combined leverages?

Earnings Per Share (EPS)

$$EPS = \frac{\text{Earnings Available to Equity Share Holders}}{\text{Number of Equity Shares Outstanding}}$$

4. A company has ₹100000, 10% debentures & 5000 equity shares outstanding. It is in the 35% tax bracket. Assuming 3 levels of EBIT

- a) ₹ 50000
- b) ₹ 30000
- c) ₹ 70000. Calculate EPS for various levels of EBIT.

Calculation of Earnings Available to Equity shareholders at different level of EBIT

Particulars	50,000	30,000	70,000
Operating Profit (EBIT)	50,000	30,000	70,000
Less:- Interest {₹1,00,000*10%}	10,000	10,000	10,000
Earnings Before Tax(EBT)	40,000	20,000	60,000
Less:- Tax @ 35%	14,000	7,000	21,000
Earnings After Tax (EAT)	26,000	13,000	39,000
Less:- Preference Dividend	----	----	----
Earnings available to equity share holders	26,000	13,000	39,000

$$EPS = \frac{\text{Earnings Available to Equity Share Holders}}{\text{Number of Equity Shares Outstanding}}$$

- (i) At EBIT ₹50,000**

$$EPS = \frac{₹26,000}{5000 \text{ shares}} = ₹5.2 \text{ Per share}$$

- (ii) At EBIT ₹30,000**

$$EPS = \frac{₹13,000}{5000 \text{ shares}} = ₹2.6 \text{ Per share}$$

- (iii) At EBIT ₹70,000**

$$EPS = \frac{₹39,000}{5000 \text{ shares}} = ₹7.8 \text{ Per share}$$

Indifference Point OR EBIT Level

Indifference point is the point of EBIT at which EPS should be same. In other words, is the point at which two different alternatives are equal.

$$\frac{(\text{EBIT} - \text{Interest}_1)(1 - t) - \text{PD}_1}{\text{No. of Equity Shares or E/S Capital} - 1} = \frac{(\text{EBIT} - \text{Interest}_2)(1 - t) - \text{PD}_2}{\text{No. of Equity Shares or E/S Capital} - 2}$$

5. Skyline software Ltd has appointed you as its finance manager. The company wants to implement a project for which ₹ 30 lakh is required to be raised from the market as a means of financing the project. The following plans & options are as follows

Particulars	Plan – A	Plan –B	Plan -C
<u>Option 1</u>			
Equity Shares	30	30	30
<u>Option 2</u>			
Equity Shares	15	20	10
12% Preference shares	Nil	10	10
10% Debentures	15	Nil	10

Assuming corporate tax to be 35% & the face value of all shares & debentures to be ₹ 100 each, calculate the indifference points & EPS for each of the financing plans. Which plan should be accepted by the company?

Solution**Calculation of indifference points & EPS for Plan -A**

⇒ **Indifference Point**

$$\frac{(\text{EBIT} - \text{Interest}_1)(1 - t) - \text{PD}_1}{\text{No. of Equity Shares or E/S Capital}} = \frac{(\text{EBIT} - \text{Interest}_2)(1 - t) - \text{PD}_2}{\text{No. of Equity Shares or E/S Capital}}$$

$$\frac{(\text{EBIT} - 0)(1 - 0.35) - 0}{₹30,00,000} = \frac{(\text{EBIT} - ₹1,50,000)(1 - 0.35) - 0}{₹15,00,000}$$

$$\frac{\text{EBIT}(0.65)}{₹30} = \frac{(\text{EBIT} - ₹1,50,000)(0.65)}{₹15}$$

$$\frac{EBIT(0.65)}{₹30} = \frac{0.65EBIT - ₹97,500}{₹15}$$

$$9.75 EBIT = 19.5 EBIT - ₹29,25,000$$

$$9.75 EBIT - 19.5 EBIT = -₹29,25,000$$

$$- 9.75 EBIT = -₹29,25,000$$

$$EBIT = \frac{₹29,25,000}{9.75} = ₹3,00,000$$

⇒ Calculation of EPS

Particulars	i	ii
Operating Profit (EBIT)	3,00,000	3,00,000
Less: - Interest	----	1,50,000
Earnings Before Tax(EBT)	3,00,000	1,50,000
Less: - Tax @ 35%	1,05,000	52,500
Earnings After Tax (EAT)	1,95,000	97,500
Less: - Preference Dividend	----	----
Earnings available to equity share holders	1,95,000	97,500
Number of Shares	30000 Sh	15000 Sh
EPS = $\frac{\text{Earnings Available to Equity Share Holders}}{\text{Number of Equity Shares Outstanding}}$	₹6.5 /Sh	₹6.5/Sh

Dividend Policy

The term dividend refers to that part of profits of a company which is distributed by the company among its shareholder. It is the reward of the shareholders for investment made by them in the shares of the company.

Determinants or Factors Influencing on Dividend Policy

1. Shareholders Expectations
2. Financial needs of the company
3. Legal Restrictions
4. Financial conditions & borrowing capacity
5. Access to capital market
6. Control

7. Competitors dividend policy
8. Liquidity/cash position
9. Past dividend rates
10. Government Control

Stable Dividend Policy (Forms of Dividend)

The stability of dividend means consistency in the stream of dividend payments. It means payment of certain minimum amount of dividend regularly. A stable dividend policy may be established in any of the following three forms:

1. Constant Dividend Per Share

Some companies follow a policy of paying fixed dividend per share irrespective of the level of earning year after year. Such firm creates reserves i.e., dividend equalization reserves to enable them to pay the fixed dividend even in the year when the earnings are not sufficient or when there are losses. The policy of constant dividend per share is more suitable to concerns whose earnings are stable over a number of years.

2. Constant Payout Ratio

It means payment of fixed percentage of net earnings as dividend every year. The amount of dividend in such policy fluctuates in direct proportion to the earnings of the company. The policy of constant payout is preferred by the firm because it is related to their ability to pay dividends.

3. Stable Rupee Dividend Plus Extra Dividend

Some companies follow a policy of paying constant low dividend per share plus extra dividend in the years of high profits such policy is more suitable to the firm having fluctuating earnings from year to year.

Stock Dividend/Bonus Share

A stock dividend represents a distribution of shares in lieu of or in addition to the cash dividend (known as bonus shares in India) to the existing shareholders. This has the effect of increasing the number of outstanding shares of the company.

In other words, whenever the company required for new investment, net profit or revenue is transferred to capital with issue of new shares to the existing shareholders on prorated basis, this process is called stock dividend or bonus shares.

Stock Split

Bonus issue is extra shares given to shareholders free of cost. Stock Split divides the existing outstanding shares of the company into multiple shares.

Similarities between bonus share & stock split

1. In both the cases, the number of shares will be increased
2. The equity shareholders ownership remains constant

Difference between bonus share & stock split

Bases	Bonus Shares/Stock dividend	Stock Split
Meaning	Bonus issue is extra shares given to shareholders free of cost.	Stock Split divides the existing outstanding shares of the company into multiple shares.
Face Value	No change	Reduces in the same ratio
Objective	To distribute accumulated earnings without paying dividends, to improve the credit worthiness of the company	To increase the affordability of stock for small retail investors, to enhance the liquidity in the market
Share capital & reserve	Share capital increases but reserve get reduces proportionately	Share capital & reserves remains the same
Future dividend per share	Remains the same since face value is same	Reduces in proportionate split ratio

